

Aggregate Supply

Unemployment and recessions (1)

- What happens during the recessions?
 - Changes in output and employment
 - Incomes fall sharply; investment goes down
 - Sometimes, problems in the financial sector (e.g., housing bubble and bank failures)
- 1930s
 - The unemployment rate in the US, which usually remained lower than 5 %, was above 10 % for more than a decade and even reached 25 % in the peak years
- 2007-09
 - Again, the unemployment rate rose sharply

Unemployment and recessions (2)

- In this chapter, we shall discuss that unemployment is a result of slow growth of aggregate demand relative to the potential output
 - Importance of the potential output: in the long-run, aggregate supply primarily depends on potential output. The growth of potential output is crucial to the long-run economic growth and improvements in the living standards
 - It is useful to distinguish between the short and the long run. In the short run, interaction of AD and AS determines the equilibrium (and recession, boom, business cycles, inflation, etc.,). On the other hand, as discussed above, in the long run, the growth of potential output is important

Potential output

- What is potential output?
 - The long-run level of production is referred to as the potential output
 - It is also known as full-employment output or the natural rate of output
 - In sum, the natural rate of output is the level of production around which the economy gravitates in the long-run
 - Potential output can change
- It is the highest sustainable level of output for an economy
 - It is not the absolute maximum that an economy can produce
 - It is possible that for some (small) period workers work over-time and the output exceeds the potential output. However, such an output level is not sustainable (for long) due to inflation: the price level rises (e.g., workers start extracting higher wages) and the unemployment rate falls
- Measurement
 - It can be measured as the output that would be produced when unemployment rate equals a benchmark level known as the non-accelerating inflation rate of unemployment (NAIRU)
 - It is the amount of output that would be produced if we remove business cycle fluctuations

Aggregate Supply (AS)

- AS: a view from the production side
 - AS curve shows the level of total national output that the producers are willing to produce and sell at each possible price level, keeping other things (e.g., monetary policy, fiscal policy, capital stock, etc.,) constant
- Shape of the aggregate supply curve
 - Unlike the aggregate demand curve, which is always downward sloping the aggregate supply curve depends crucially on the time horizon
 - Short-run: AS curve is upward sloping
 - The fact that higher output is supplied at higher prices seems to be the usual behaviour of these two variables
 - In the SR, (some components of) wages and prices are rigid or inflexible. As a result, when AD rises, the businesses can earn more profits (since the costs they are facing do not change) and respond to this situation by supplying more output
 - Long-run: AS curve is vertical
 - The sticky components of cost become unstuck and negotiable. As the **prices rise, labor recognizes that prices have risen and ask for increases in wages. Ultimately, all costs adjust to the higher output prices**
 - Once costs are up, the firms would no longer benefit from the higher level of AD and have no incentive to supply more. The **AS curve is vertical and is determined by the potential output**

Aggregate supply in the long-run

- Long-run AS is primarily determined by potential output
- Potential output: depends on
 - Inputs: labor, capital, and natural resources
 - The potential output is achieved when “employment of labor and other inputs is at the maximum sustainable level”
 - Technology (and efficiency): transforms the inputs into output
- The level of output (note that AS curve is vertical) is the potential output or the natural rate of output
- The proponents of this view are known as *classical* macroeconomists
- A change in the aggregate demand affect the prices but has no effect on the real output
 - The prices and wages “adjust fully to changes in aggregate demand”

Price
Level

P_1

P_2

Long-run
aggregate
supply

1. A change
in the price
level . . .

2. . . does not affect
the quantity of goods
and services supplied
in the long run.

0

Natural rate
of output

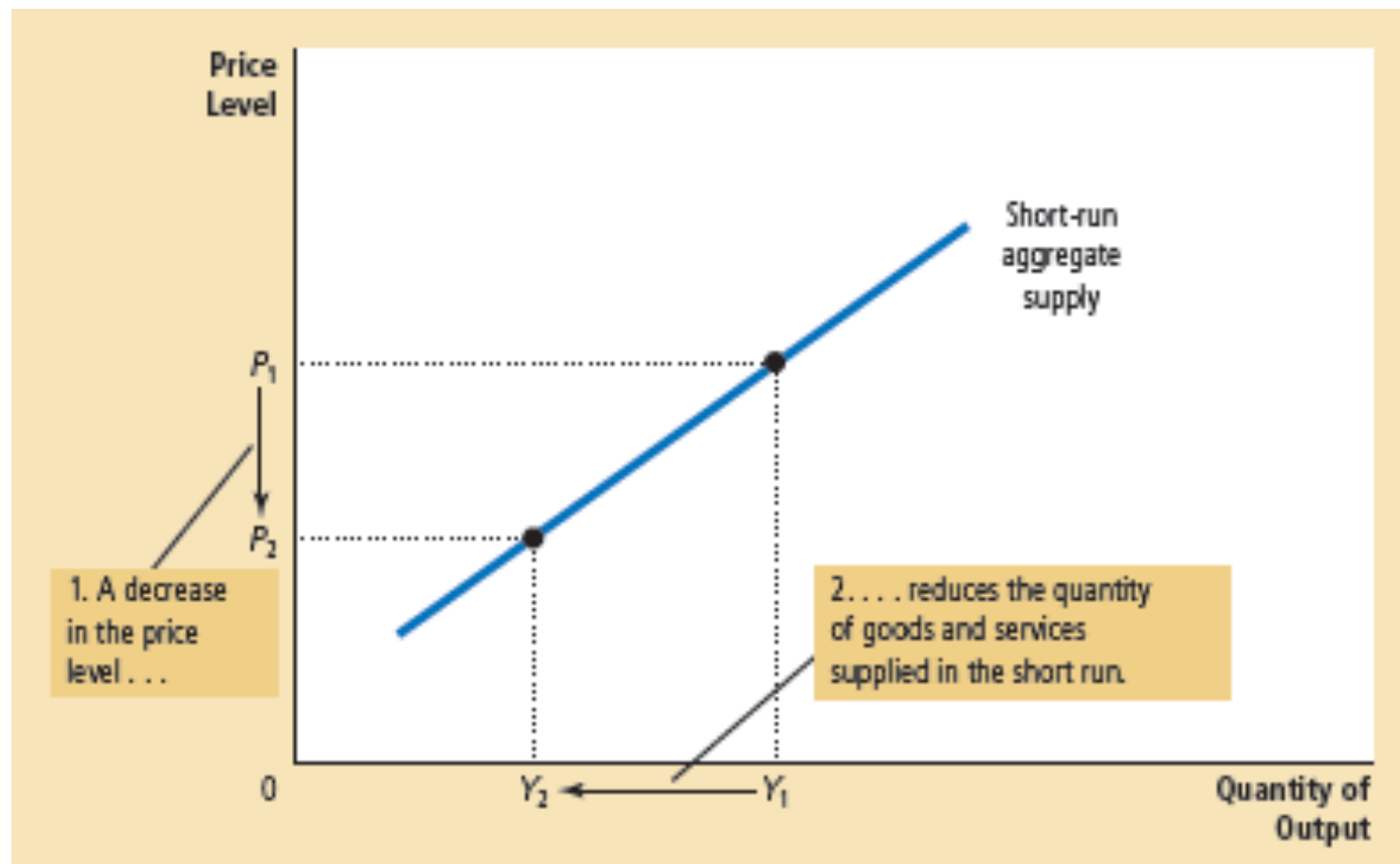
Quantity of
Output

Shifts in long-run AS

- Potential output is not a static concept
 - As an economy grows, so is its potential output
- Shifts in long-run AS
 - Labor: immigration
 - Changes in capital stock
 - Natural resources: events in world oil market have shifted AS curves; discovery of new resources, climate change
 - Technology: advent of computers

Aggregate supply in the short-run

- In the long-run, the overall level of prices does not affect an economy's ability to produce goods and services but in the short-run, it does. In the short-run, AS curve is upward sloping
 - This view is known as *Keynesian* macroeconomics
 - A change in the aggregate demand affects both the prices and the real output
- We shall next have a look to a theory explaining why do changes in the price level affect the AS curve in short-run. The central theme underlying the theory is
 - The actual output deviates from the long-run output when the actual price level deviates from the expected price level



Sticky-wage theory

- *Sticky* wages in the short-run: Nominal wages are slow to adjust to changing economic conditions
 - Long-term contracts between workers and firms that fix nominal wages
 - Wage rigidity and informal sector
- Nominal wages are based on the expected price level and do not respond immediately when the actual price level turns out to be different
- And, the stickiness of wages means firms produce less (more) output when the price level turns out lower (higher) than expected (see, [example](#) next slide)

Sticky wages – example

Mankiw (p. 736)

aggregate-supply curve that slopes upward. Imagine that a year ago a firm expected the price level today to be 100, and based on this expectation, it signed a contract with its workers agreeing to pay them, say, \$20 an hour. In fact, the price level, P , turns out to be only 95. Because prices have fallen below expectations, the firm gets 5 percent less than expected for each unit of its product that it sells. The cost of labor used to make the output, however, is stuck at \$20 per hour. Production is now less profitable, so the firm hires fewer workers and reduces the quantity of output supplied. Over time, the labor contract will expire, and the firm can renegotiate with its workers for a lower wage (which they may accept because prices are lower), but in the meantime, employment and production will remain below their long-run levels.

The same logic works in reverse. Suppose the price level turns out to be 105, and the wage remains stuck at \$20. The firm sees that the amount it is paid for each unit sold is up by 5 percent, while its labor costs are not. In response, it hires more workers and increases the quantity supplied. Eventually, the workers

Sticky-price theory

- The prices of some goods and services also adjust sluggishly in response to changing economic conditions
 - Price adjustment is slow because there are costs to adjusting prices, called menu costs
 - Menu costs include costs of printing and distributing catalogs and the time required to change price tags
 - Rental agreements are signed in monetary unit terms (not linked to the inflation rates)
- Note that over time nominal wages and prices will become unstuck and in the long-run, wages and prices are flexible

Sticky prices – example

Mankiw (p. 737)

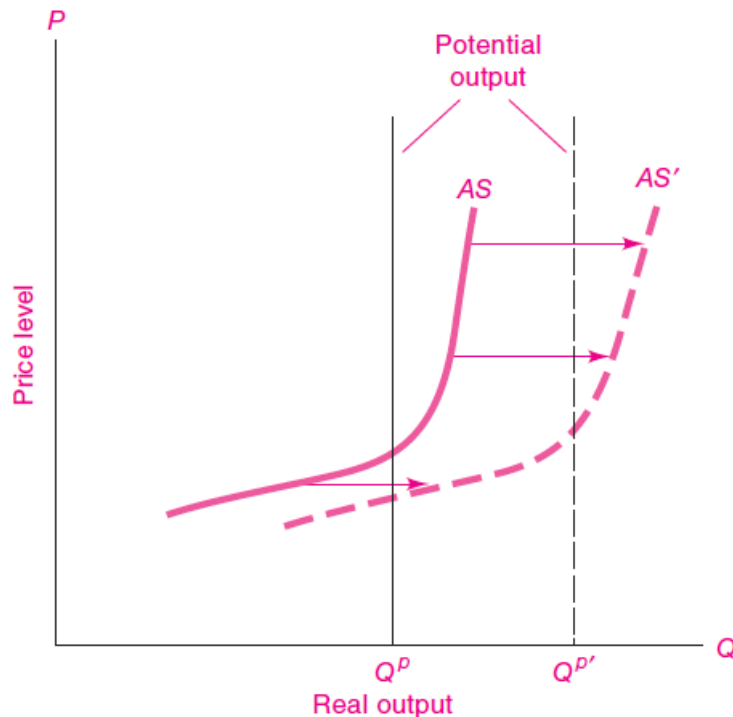
To see how sticky prices explain the aggregate-supply curve's upward slope, suppose that each firm in the economy announces its prices in advance based on the economic conditions it expects to prevail over the coming year. Suppose further that after prices are announced, the economy experiences an unexpected contraction in the money supply, which (as we have learned) will reduce the overall price level in the long run. Although some firms can reduce their prices immediately in response to an unexpected change in economic conditions, other firms may not want to incur additional menu costs. As a result, they may temporarily lag behind in reducing their prices. Because these lagging firms have prices that are too high, their sales decline. Declining sales, in turn, cause these firms to cut back on production and employment. In other words, because not all prices adjust instantly to changing economic conditions, an unexpected fall in the price level leaves some firms with higher-than-desired prices, and these higher-than-desired prices depress sales and induce firms to reduce the quantity of goods and services they produce.

Shifts in short-run AS (1)

- Short-run AS shifts due to the following factors
 - Inputs: labor, capital, and natural resources
 - A reduction in input costs, an increase in capital stock
 - Technology (and efficiency)
 - An improvement in technology
 - *Expected* price level (to account for the theories that explain the short-run behavior)
- The output produced in the short-run is also affected by sticky wages and prices. Wages and prices are set based on the expected price level. When people change their expectations of the price level, the short-run AS curve shifts
 - When the *expected* price level rises, workers bargain for high nominal wages; with higher wages, firms' costs increase, and firms produce a smaller quantity of goods and services [In other words, the effect of higher P^e is similar to that of the increase in input costs]
 - Thus, an increase in the expected price level reduces the quantity of goods and services supplied and shifts the short-run aggregate-supply curve to the left

Shifts in short-run AS (2)

(a) Increase in Potential Output



(b) Increase in Costs

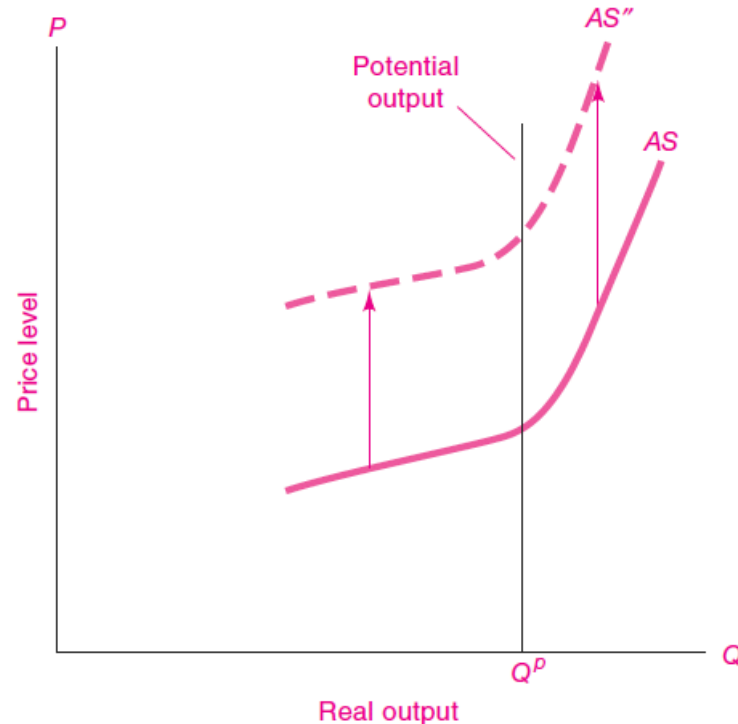
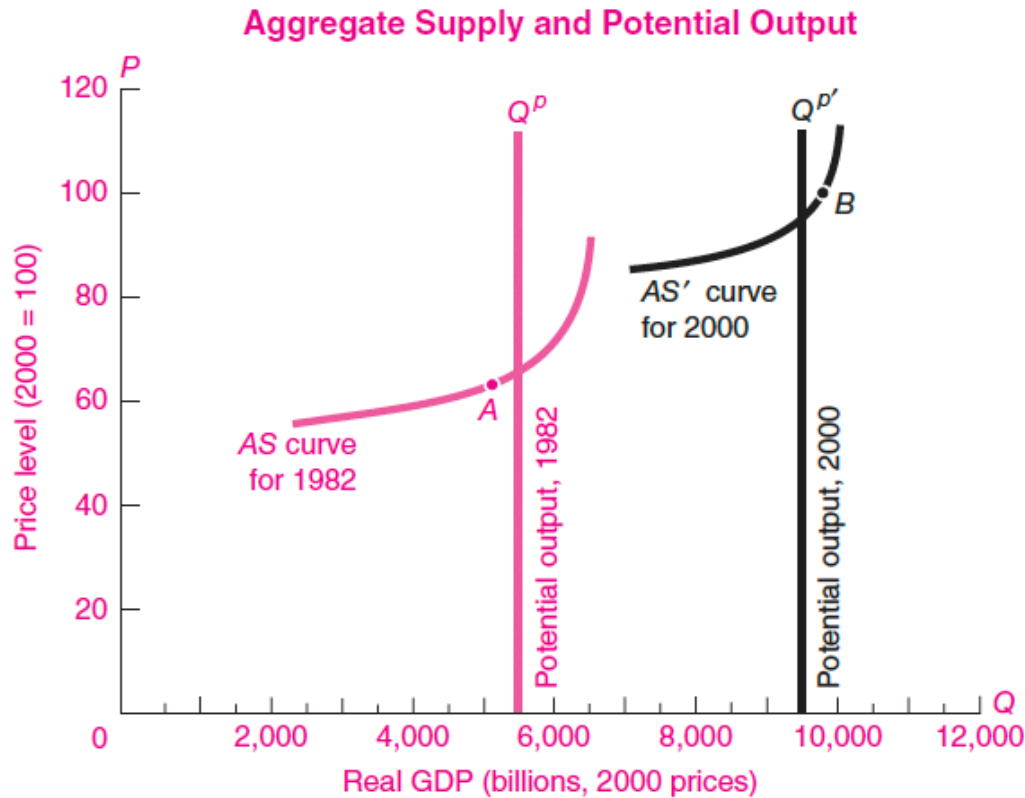


FIGURE 29-1. How Do Growth in Potential Output and Cost Increases Affect Aggregate Supply?

In (a), growth in potential output with unchanged production costs shifts the AS curve rightward from AS to AS'. When production costs increase, say, because of higher wages or oil costs, but with unchanged potential output, the AS curve shifts vertically upward, as from AS to AS'' in (b).

Shifting of AS: a real-world case



Samuelson:

“real potential output grew about 72 percent” between 1982 and 2000” due to “due to increases in capital and labor inputs along with technological improvements”

Simultaneously, there was an increase in costs (wages, oil prices, etc.,) which had led to the upward shift

Aggregate demand and supply

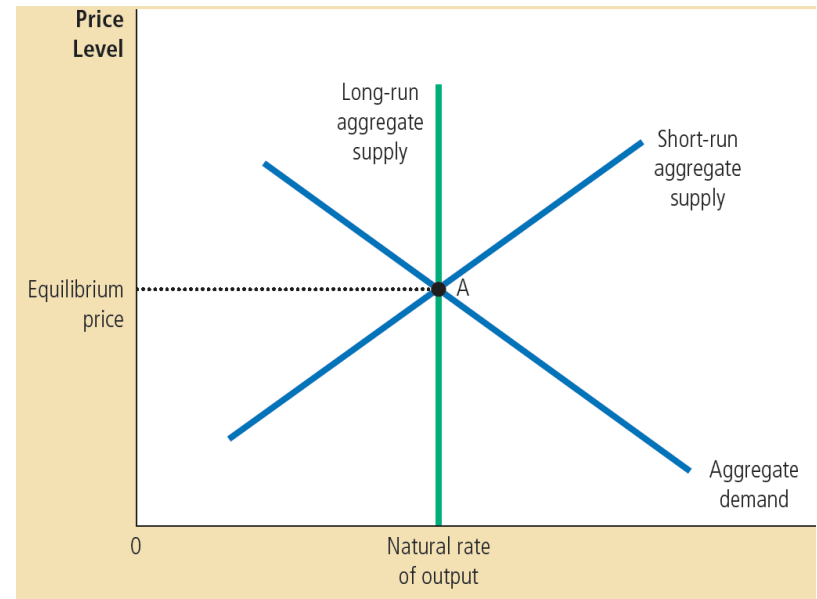
- Short-run
 - Changes in aggregate demand significantly affect the output. If, for instance, a monetary tightening leads to fall in the demand, output and the price level will fall
 - Therefore, in the short-run, the Keynesian approach for economic stabilization that focuses on changes in AD is helpful
- Long-run
 - Changes in aggregate demand do not have any impact on the real output, which in turn is determined by potential output. It only affects the price level
 - In the long-run, the classical approach is to focus on the potential output
- Long-run equilibrium is achieved when aggregate demand curve crosses the long-run AS curve. At this point, $P^e = P$

The long run equilibrium

- While analyzing the impact of a change on the equilibrium, we assume that the economy is in long-run equilibrium
- The equilibrium is determined when AD intersects LRAS
- In the equilibrium,

$$Y = \bar{Y}, P = P^e$$

- Because economy is always in SR equilibrium, SRAS will also pass through the equilibrium



Examining macroeconomic fluctuations

- How does an event affect the macroeconomic variables?
- Four steps:
 - Determine whether the event shifts the AD curve or the AS curve or both
 - Decide the direction of shift
 - Determine the short-run impact using the AD-AS framework
 - Analyze the movement from short-run to long-run

Shift in AD

- Consider a wave of pessimism in the economy
 - Households and business are worried about the future and cut back their spending/ investment. Since spending is affected, the AD curve is affected
 - Households and firms want to buy smaller quantities, it reduces AD (AD shifts to left)
 - Short-run: Output and the price level fall
 - As the price level is lower, the price expectations would adjust. Specifically, the people would revise the price expectations downwards
 - Note that the AS curve in the short run (SRAS) is drawn, given the price expectations. Thus, the economy would be on another curve consistent with the new P^e
 - Long-run: Output is back to the long-run level, but the price level is low
 - Role of policy: policymakers can attempt to shift the AD back to its original position (at the time of initial shift itself)

Shift in AS

- Consider bad weather (unseasonal rains)
 - Consequences
 - Cost of production increases; loss of crop output
 - Since supply is affected, the AS curve would shift
 - Which direction? the production costs are higher, firms supply lower quantities, it reduces AS (AS shifts to left)
 - Short-run:
 - Output falls and the price level rises
 - This condition is known as stagflation (stagnation + inflation)
 - In fact, as the price level rises, workers may demand higher wages. P^e may rise, this will again lead to a further leftward shift of the SRAS curve, making stagflation even worse. This is referred to as the wage-price spiral
 - The spiral will slow. As the employment is low, it will put downward pressure on wages (and the SRAS may shift rightward)
 - Long-run:
 - Policymakers act
 - Output is back to the long-run level, but the price level is up

References

- Aggregate Supply
 - Chapter 33: Mankiw
 - Chapter 29: Samuelson

Unemployment

Impact of unemployment

- High unemployment: economic and social problems
 - Social problem:
 - people suffer – human and psychological toll; reduction in income; demographic consequences (next slide)
 - Macroeconomic problem: waste of valuable resources
 - Economy is throwing away the goods and services that the unemployed people could have produced
 - Consider, for instance, the opportunity cost of recessions

Incidence	Avg. unemp. rate (%)	Loss in GDP (% of avg. GDP)
Great Depression (1930-39)	18.2	30
Oil and inflation crisis (1975-84)	7.7	2.7
Slump in 2000s (2001-03)	5.5	1.4



Employment Instability and Childbirth over the Last 20 Years in Italy

Stefani Scherer¹  · Elisa Brini² 

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Abstract

Family formation is fostered by circumstances of plannability and economic and social stability. Conversely, as documented in previous literature, employment instability can hamper fertility decisions. Based on data from the Italian Labour Force Survey, this paper examines the association between employment-related instability and the likelihood of having a first or additional child from 2000 to 2020 in Italy, covering a period characterised by increasing labour market deregulation. Our results show that individual employment instability, such as temporary employment or unemployment, negatively influences the likelihood of having a first and second child, while the progression to higher parities is less affected by employment situations. Building upon previous research, we demonstrate how the negative associa-

Okun's Law (1)

- One of the worst consequences of a recession is rise in unemployment
- Reduction in output -> firms need fewer labor -> no new hiring and existing workers are laid off
- Thus, there is an inverse relationship between the changes in output and the unemployment rate
 - Empirical evidence
 - For every 2 percent fall in GDP relative to the potential GDP, the unemployment rate rises by 1 percentage point
 - If GDP begins at 100 per cent of its potential and falls to 98 per cent of the potential, the unemployment rate rises by 1 percentage point

Okun's Law (2)

- Consequences
 - Actual GDP must grow as rapidly as the potential GDP in order to prevent the unemployment rate from rising
 - In other words, it is possible to reduce unemployment rate by growing at a rate higher than that of the potential GDP
- Note that the quantification holds for the short-run

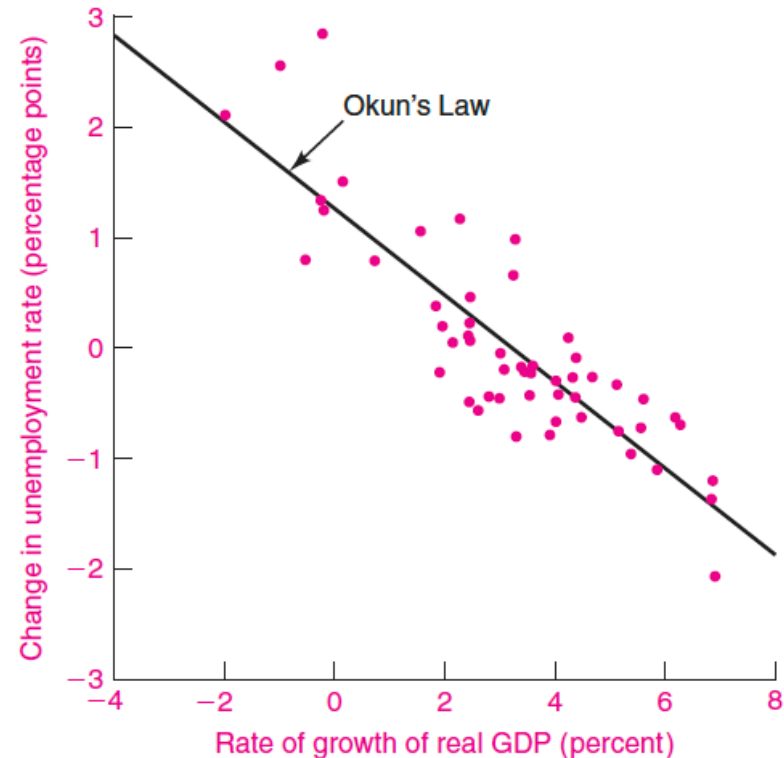
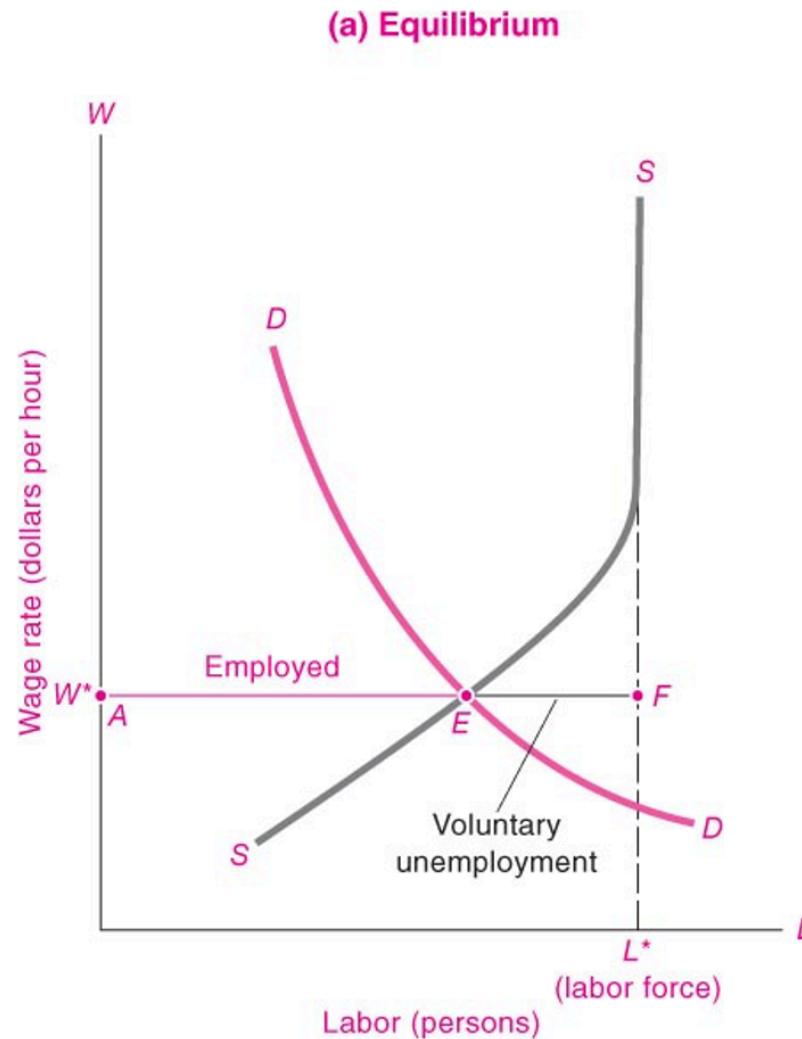


FIGURE 29-5. Okun's Law Illustrated, 1955–2007

Equilibrium unemployment (1)

- Is the unemployment which prevails when the economy is in equilibrium (i.e., the markets clear)
- Equilibrium unemployment arises when people become unemployed voluntarily as they move
 - from job to job or
 - into and out of the labor force
- Is also known as *frictional or voluntary* unemployment
 - Frictional: people cannot move instantaneously between the jobs
- This is equilibrium unemployment as firms and workers are on their demand and supply schedules (next slide)
 - All those who want a job at the prevailing wage rates and market conditions are employed
 - At the prevailing wage rate, the utility derived from unemployment (e.g., leisure or an unpaid maternity leave) is higher than that from working
- There may be people who would like to work (EF in the graph) but only at a higher wage rate
 - They have chosen to be unemployed at the prevailing market wage rate
 - Despite choosing to be unemployed, they would be counted as unemployed in the official statistics

Equilibrium unemployment (2)



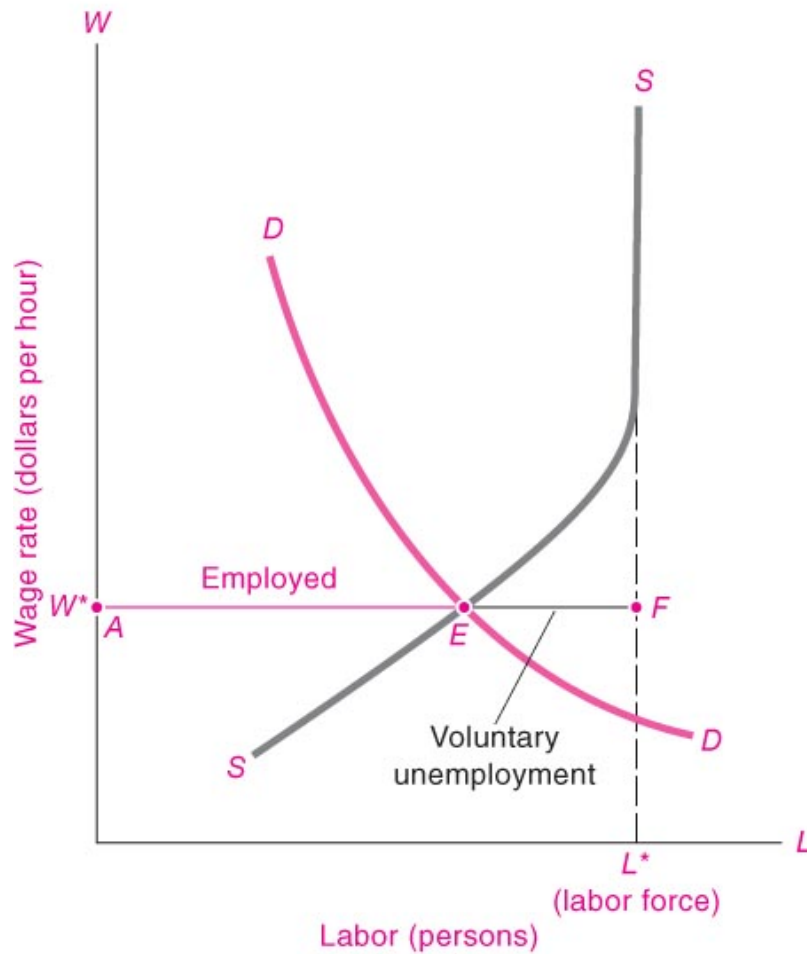
Disequilibrium unemployment (1)

- The labor market may not always function properly, or there may be a macroeconomic issue
 - Even the people who want a job at the prevailing wage rates may not find jobs
- This is known as disequilibrium unemployment and can arise because of two reasons
- Structural unemployment: mismatch in the demand and supply of workers
 - Mismatch across the occupations
 - The demand for some types of labor is high whereas for others, it is low. Markets may not adjust quickly
 - Mismatch across regions
 - Punjab during crop harvesting season and some other states

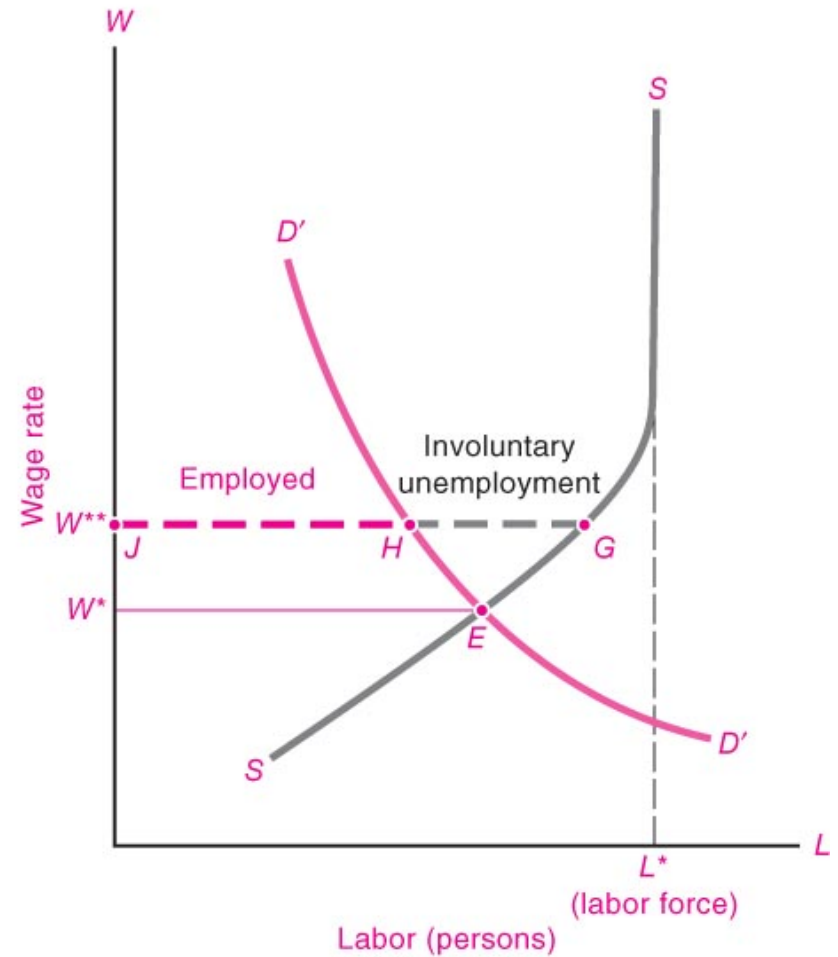
Disequilibrium unemployment (2)

- Cyclical unemployment
 - Overall demand for labor responds to the business cycles
 - 2007-09: demand for labor declined and unemployment rates increased in most of the industries and regions
 - 2000s expansion: unemployment rates fell in the US
 - The impact of business cycles on unemployment may vary
- Disequilibrium employment: labor markets are not at the demand – supply equilibrium
 - Next slide; wages are at W^{**} in SR and are sticky; do not adjust to the equilibrium wage that should prevail
 - As the labor demand declines to $D'D'$, the market wage rate is still at W^{**} (i.e., $>$ the equilibrium level, W^*)
 - HG is disequilibrium unemployment
 - Note that this is involuntary unemployment; even those who wants a job at W^{**} may not find jobs

(a) Equilibrium



(b) Disequilibrium

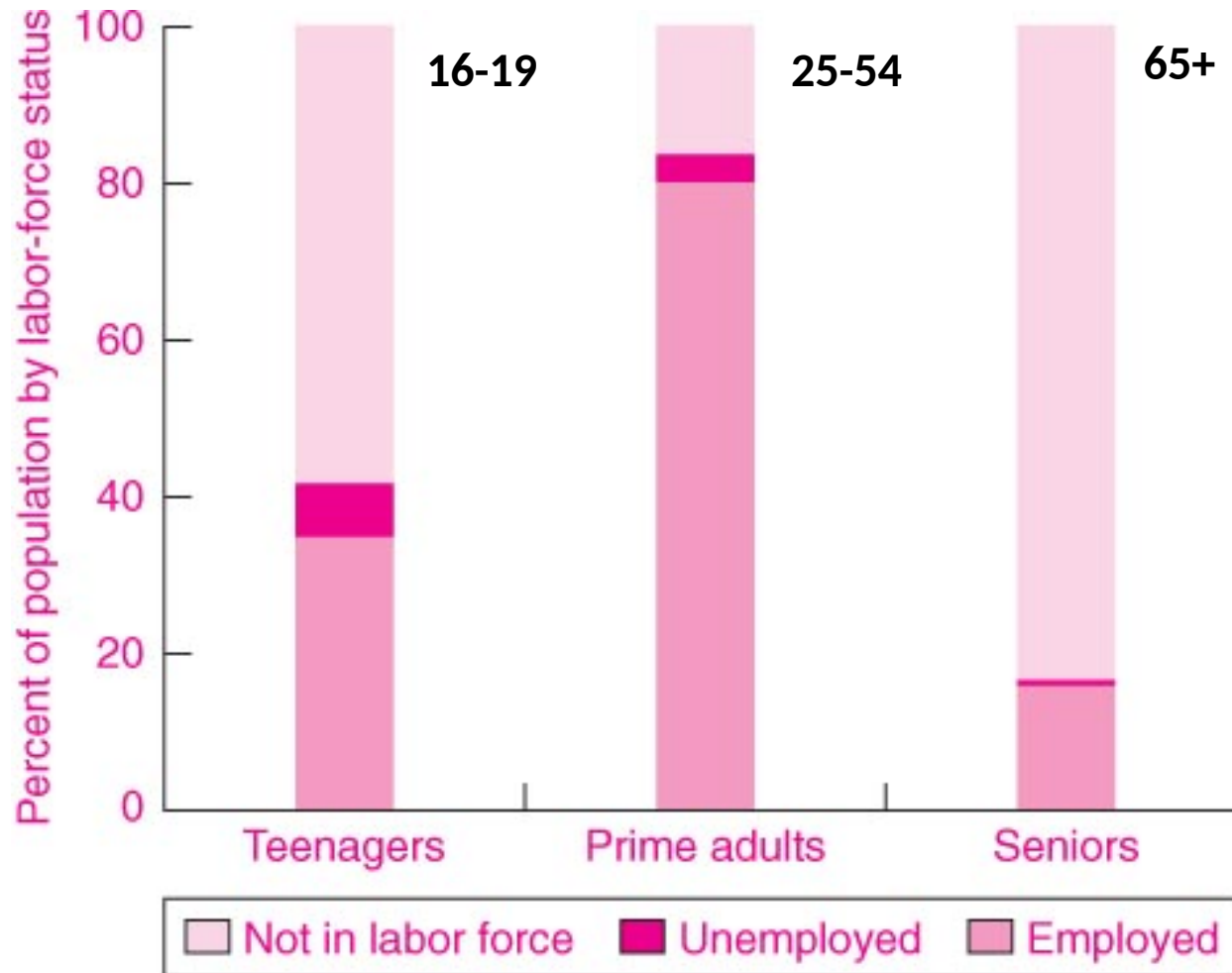


Measurement of unemployment

- The population can be divided into the following (mutually exclusive and exhaustive?) categories
 - Employed (E) – having any paid job, self-employed or unpaid work in family business; includes both full- and part-time
 - Unemployed (U) – available for job but are not employed
 - Not in the labor force (N) – kids, students, home-makers, retirees
- Important Indicators
 - Labor force = $E + U$
 - Unemployment rate (UR) = $U * 100 / (E + U)$
 - Labor force participation rate (LFPR) = $(E + U) * 100 / (E + U + N)$
 - Worker population ratio (WPR) = $E * 100 / (E + U + N)$

Structure of the US population

(above 16 years of age; 2007)



PLFS 2023-24

LFPR estimates

15-29 years: 46.5

Above 15 years : 60.1

Male: 58.2

Female: 31.7

Rural: 46.8

Urban: 41.0

(Urban female is lower than rural female)

Unemployment rate

All: 3.2

Male: 3.2

Female: 3.2

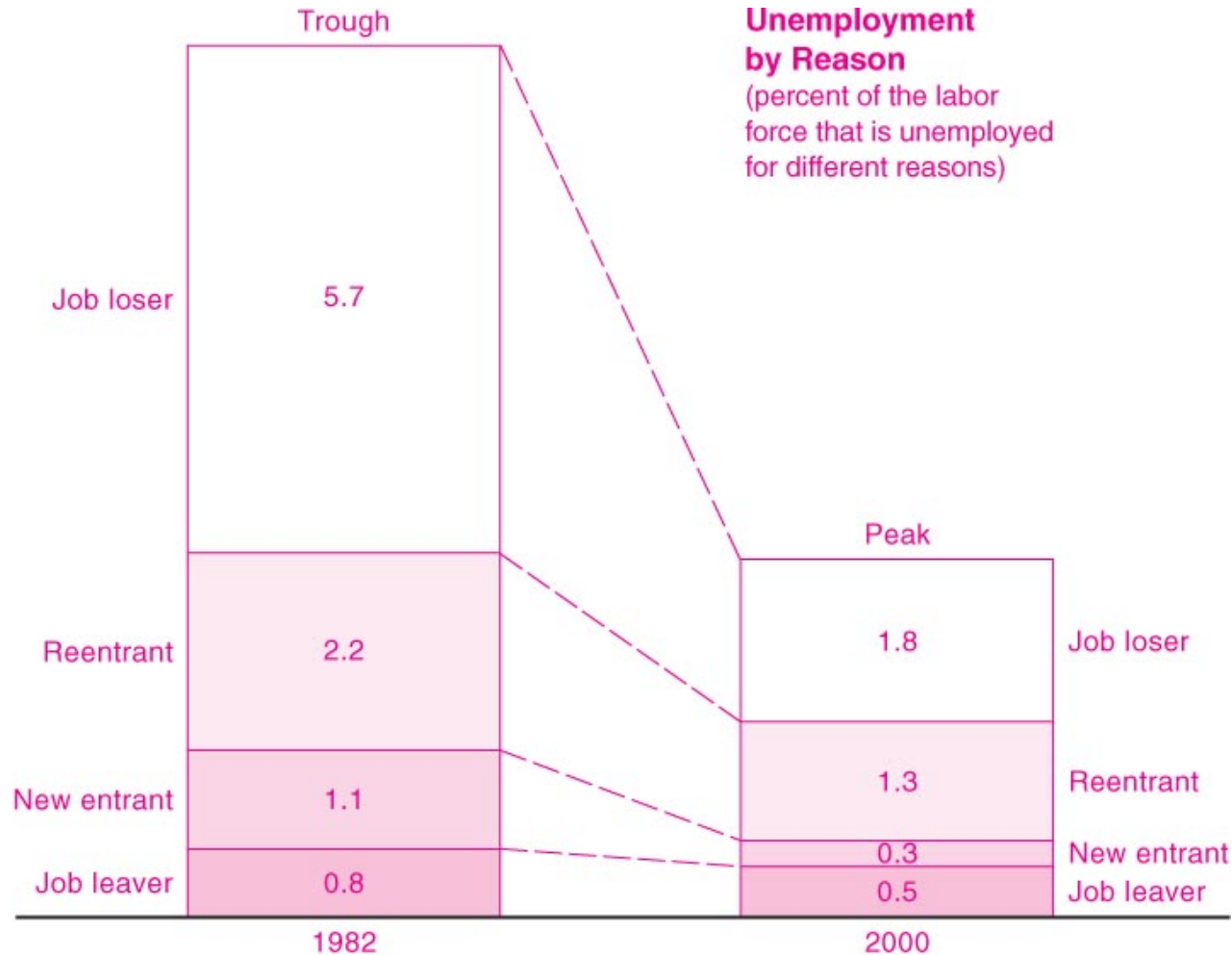
Rural: 2.5

Urban: 5.1

15-29 years: 10.2

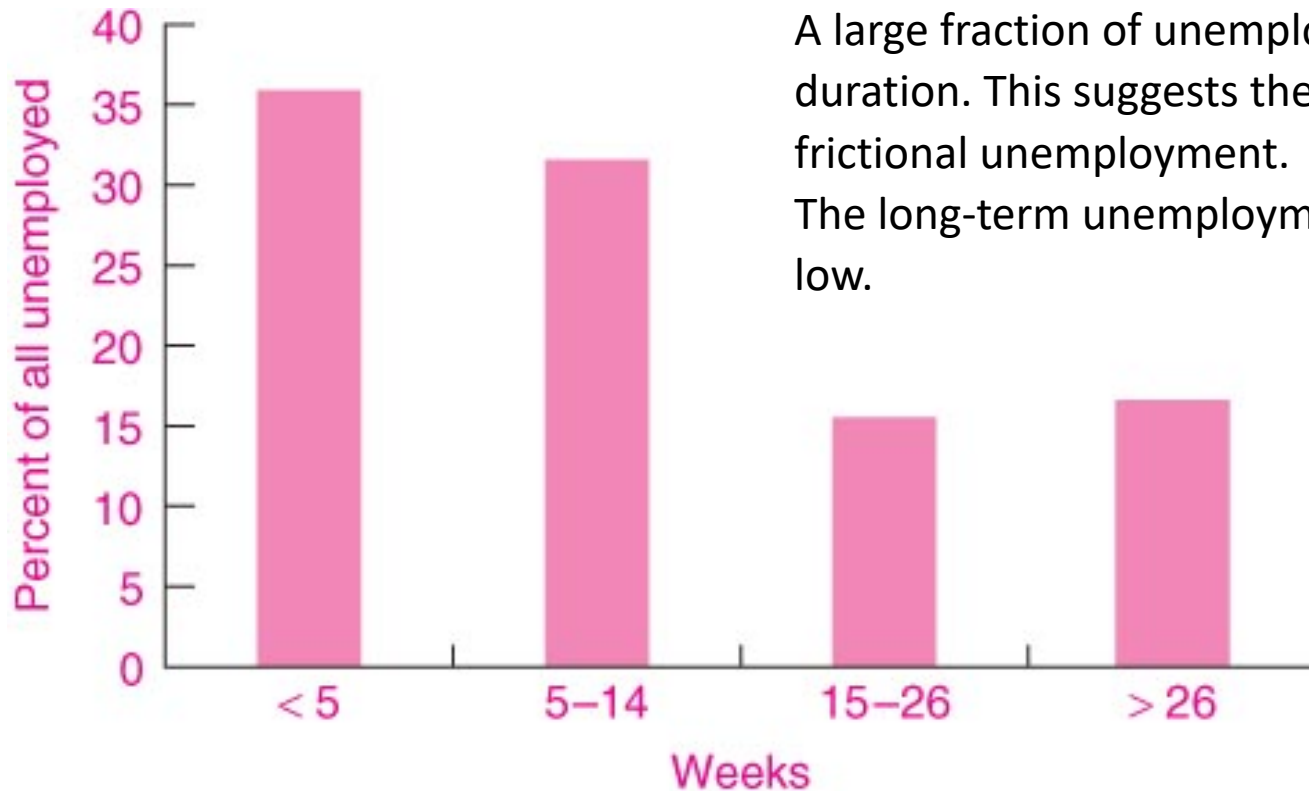
Why are people unemployed?

[unemployment rates: 9.7 % (1982) and 4.1 % (2000)]



Duration of unemployment

the United States, 2000-2007



A large fraction of unemployment is of short duration. This suggests the presence of frictional unemployment. The long-term unemployment is relatively low.

Informal employment



Dr. Sandeep Singh was a professor in Punjab. Punjab Man With PhD, 4 Master's Degrees Sells Vegetables To Make Ends Meet

"I had to leave the job because I didn't use to get my salary on time and there was frequent pay cut. It had become difficult for me to make ends meet with that job. That's why I switched to selling vegetables for the survival of me and my family." With his vegetable cart and a board saying "PhD Sabzi Wala", Dr. Sandeep Singh goes home to home every day to sell vegetables. He says he earns more money by selling vegetables than he did as a professor."

Source: <https://www.ndtv.com/india-news/punjab-man-with-phd-4-masters-degrees-sells-vegetables-to-make-ends-meet-4776202>

Indian labor market (1)

(Source: Nagaraj 2007)

- Size: 430 million workers in 2004-05
 - Growth: 2 per cent per annum
 - A stable worker-population ratio of around 40 per cent
- Women's participation in workforce
 - was 28 per cent
 - Likely to be an underestimate
- Child labor
 - 13 million in 2001
 - share in workforce declining
- Rural workers constitute over 60% of the workforce

Indian labor market (2)

(Source: Nagaraj 2007)

- Organised sector
 - Employs 8% of the workforce
 - Producing 40% of GDP
- Labor legislations
 - Mostly deal with the organised sector
 - No economy-wide social security
 - Labor being a concurrent subject in the constitution, states are empowered to enact separate legislations
- Structural transformation
 - Agriculture's share declining from 68.5% in 1983, to 55.4% in 2004-05
 - But the workforce is shifting more towards services than to industry

Policies to reduce unemployment

- High unemployment is costly for the society and hence should be reduced
- The following measures can help reducing unemployment
 - Improving labor market services
 - Frictional and structural unemployment can be reduced by improving access to information
 - Training programs
 - Preparing unskilled workers for job by imparting skills
 - Reducing disincentives to work
 - Reforming unemployment-benefits (relevant for developed countries)

Skill mismatch

S. No.	Field of training	% matched
1	Mechanical engineering trades	63.53
2	Electrical and electronic engineering trades	55.05
3	Computer trades	72.83
4	Civil engineering- and building construction-related works	74.23
5	Chemical engineering trades	64.55
6	Leather-related work	31.90
7	Textile-related work	65.34
8	Catering, nutrition, hotels and restaurant-related work	51.15
9	Artisan/ craftsman/ handicraft and cottage-based production work	59.17
10	Creative arts/ artists	59.11
11	Agriculture and crop production-related skills and food preservation related work	82.76
12	Non-crop-based agricultural and other related activities	58.63
13	Health and paramedical services-related work	75.92
14	Office and business-related work	62.62
15	Driving and motor mechanic work	63.82
16	Beautician, hairdressing and related work	71.45
17	Work related to tour operators/ travel managers	82.82
18	Photography and related work	86.26
19	Work related to childcare, nutrition, pre-schools and crèche	87.38
20	Journalism, mass communication and media-related work	61.00
21	Printing technology-related work	77.45
22	Other	–
	Total	66.36

Indian labor market (3)

(Source: Nagaraj 2007)

	1983		1993-94		2004-05	
	% GDP	% Workforce	% GDP	% Workforce	% GDP	% Workforce
Agriculture	40.0	68.5	30.0	64.0	20.2	56.5
Industry	24.3	13.8	25.2	14.9	26.1	18.8
Services	35.7	17.6	48.8	21.1	53.7	24.8

Current composition (PLFS 2023-24: p. 15): 46 (Agri), 12 (Manu), 42 (Services)